

# ED-FEED: An AI-Driven Intelligent Student Feedback Analysis and Insight Generation System

## Implementation of Baseline Model

The baseline model was developed to establish a reference performance for sentiment classification on the educational feedback dataset. The dataset consisted of labeled comments categorized into negative, neutral, and positive sentiments.

For feature extraction, **TF-IDF vectorization** was used with a limited number of features and inclusion of n-grams to capture both individual words and contextual word patterns. This approach helped convert textual data into a structured numerical format suitable for machine learning models.

The model implemented for baseline evaluation was **Logistic Regression**, chosen for its simplicity and effectiveness in text classification tasks. The dataset was preprocessed and split into training and testing sets to ensure proper evaluation.

During evaluation, the model achieved an overall accuracy of approximately **82%**, indicating strong general performance. The model performed well in predicting positive and negative sentiments, showing high precision and recall for these classes.

However, a key limitation observed was poor performance in detecting **neutral sentiment**, with significantly low recall. This issue is primarily due to class imbalance and the inability of traditional machine learning models to capture deeper contextual meaning in text.

In conclusion, the baseline model provides a solid starting point for sentiment classification, but it lacks the capability to effectively handle contextual nuances in language. This highlighted the need for more advanced models to improve performance.

## RESULT

### Baseline Model Performance

The baseline models were evaluated using both classification and regression approaches to understand overall performance.

#### Sentiment Classification (Logistic Regression)

The classification model achieved an overall accuracy of approximately **77%**, indicating good general performance on the dataset.

## Classification Report

| Class    | Precision | Recall | F1-Score | Support |
|----------|-----------|--------|----------|---------|
| Negative | 0.80      | 0.72   | 0.76     | 1209    |
| Neutral  | 0.36      | 0.10   | 0.16     | 478     |
| Positive | 0.78      | 0.94   | 0.85     | 2308    |

Overall Accuracy: **0.77**

Macro Avg F1: **0.59**

Weighted Avg F1: **0.74**

The model performs well in identifying **positive and negative sentiments**, but shows poor performance for the **neutral class**, with very low recall. This indicates difficulty in detecting neutral sentiment, likely due to class imbalance and limited contextual understanding.

## Regression-Based Models

To further evaluate performance, regression models were also tested:

- **Linear Regression**
  - Mean Squared Error (MSE): **0.192**
  - R<sup>2</sup> Score: **0.91**
- **Random Forest**
  - Mean Squared Error (MSE): **0.225**
  - R<sup>2</sup> Score: **0.89**

These results indicate that Linear Regression slightly outperforms Random Forest in terms of prediction accuracy for sentiment scoring.

## Confusion Matrix Analysis

The confusion matrix shows that most predictions are correctly classified along the diagonal, particularly for the positive class. However, there is noticeable misclassification in the neutral category, where many samples are incorrectly predicted as positive or negative.

## Conclusion

The baseline models demonstrate strong overall performance, particularly for positive sentiment classification. However, they struggle with neutral sentiment detection due to limited contextual understanding. This highlights the need for more advanced models, such as deep learning approaches, to improve performance across all classes.